

Al Muqshith Shifan

Applied ML / RL Developer • Systems Engineering • Research Assistant

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PROFESSIONAL SUMMARY

MSc Computer Science candidate specializing in **software design for artificial intelligence and reinforcement learning (RL)** with hands-on experience building DRL training infrastructure (multi-head PPO critics, custom PyTorch environments, persona-based reward systems) and programmable-network research systems on **Intel Tofino P4** silicon. CCNA-certified; comfortable in low-level Linux/Unix stacks. Looking to contribute to GPU-native ML tooling.

EDUCATION

Ontario Tech University

MSc in Computer Science – Software Design (AI/RL Focus)

Oshawa, ON

May 2026 – Present

Ontario Tech University

Bachelor of IT – Networking & Cybersecurity; GPA 3.8/4.2 (Dean's List 2026)

Oshawa, ON

Graduated Jun 2026

PUBLICATIONS

K. C. Chua, **A. M. M. Shifan**, A. Neshati, L. Zaman, and C. Politowski, "RIDGE: State-Conditioned Reward Blending for Behavioral Coverage in Deep RL Game Agents," *IEEE Conference on Games (CoG)*, Madrid, Spain, Sept. 2026. **Accepted.**

WORK EXPERIENCE

Graduate Researcher – SEGAL Lab

May 2026 – Present

Ontario Tech University

Oshawa, ON

- Research on deep reinforcement learning for game agents and emotion-grounded memory for LLM-driven NPCs; co-authored work accepted at **IEEE CoG 2026**.
- Own the full research stack end to end: PyTorch training infrastructure, reproducible experiment sweeps, evaluation harnesses with cross-validated tuning and permutation-based significance testing, and the papers themselves.

Graduate Researcher – Code&SorceryLab

2025 – Present

Research collective (with SEGAL Lab collaborators)

Oshawa, ON

- Maintain the group's public research codebase – RIDGE, PEAK, RANGE, Procedural-Graph-Tools and the Steam patch-notes dataset – with reproducible training pipelines, pinned configs and documented experiment protocols.
- Set the engineering standards the research runs on: deterministic seeding, versioned experiment sweeps, and evaluation harnesses written before results are claimed.

Teaching Assistant – Networking I & Advanced Networking II

Sep 2025 – Present

Ontario Tech University

Oshawa, ON

- Led weekly lab sessions for ~120 undergraduates across two networking courses, guiding hands-on Cisco IOS configuration of routers, switches, and stateful firewalls; mentored students through TCP/IP, VLSM subnetting, and routing protocol debugging in office hours.

Network Administrator – Summer Research Assistant

Jun 2025 – Sep 2025

Ontario Tech University

Oshawa, ON

- Engineered programmable-network research infrastructure on **Intel Tofino silicon P4 switches**, supporting faculty research on in-network telemetry and traffic analysis.
- Authored custom P4 data-plane programs extracting per-flow header and timing telemetry at line rate, delivering the lab's first working Tofino-deployed implementation.
- Diagnosed and patched recurring Linux kernel and driver issues across P4 control-plane servers, reducing reported system crashes by ~30% over a 3-month window.

Cyber Security Analyst

Jun 2025 – Sep 2025

YYC Beeswax Ltd

Remote

- Performed vulnerability assessments on a WooCommerce + Stripe e-commerce stack, identifying 15 exploitable issues across customer-facing endpoints with prioritized remediation steps.
- Audited 30+ WordPress/WooCommerce plugins against CVE databases and authored a hardening checklist adopted as the engineering team's pre-deploy review gate.

Networking Lab Technician

Sep 2024 – Jun 2025

Ontario Tech University

Oshawa, ON

- Cut lab network downtime by ~20% for 50+ daily users by isolating switching bottlenecks and reconfiguring VLAN trunking; maintained 20+ workstations and DHCP/DNS/NTP services, resolving Tier-2 tickets.

PROJECTS

RIDGE – Reactive Inter-persona Dynamic Goal Engine | Python, PyTorch, Crafter, PPO

2026

- Second author on a five-author deep RL architecture for state-conditioned reward blending across four persona objectives (Explorer, Survivor, Craftsman, Warrior) on Crafter. **Accepted at IEEE Conference on Games (CoG) 2026**, Madrid, Spain. GitHub: Code-SorceryLab/RIDGE.
- Designed a **multi-head PPO critic** – one value head per persona on a shared encoder, each trained on its own stationary reward – paired with a state-conditioned blender producing softmax-normalized sigmoid weights, addressing the switching-stability dilemma that breaks PPO under non-stationary rewards.
- Result: a single agent acquires both branches of Crafter's wood-tier crafting/combat split (0.45 pickaxe, 0.51 sword) that no fixed-persona baseline solves jointly, with value loss held at ≈ 0.04 – matching the most stable specialist. Accepted as a work-in-progress paper; full multi-seed validation planned.
- Built the full training pipeline in PyTorch with TensorBoard logging and YAML configs; ran the 1M-step Crafter budget across four fixed-persona baselines and a five-point sharpness ablation, concentrating seeds ($n=3$) on the central conditions.

Emotional Memory in LLM-Driven NPCs | Systematic mapping study; in preparation, IEEE ToG

2026

- Sole-reviewer systematic mapping study of emotional memory in LLM-driven game characters: screened 66 records against pre-registered inclusion criteria, coded the 30 included studies across six dimensions, and executed formally verified search strings against Scopus, IEEE Xplore, and the ACM Digital Library.
- Central finding is a **regression, not a gap**: pre-LLM architectures (FLAME, FAtiMA, Prom Week) coupled emotion to memory, but the field-standard memory stream dropped the emotion term and only 3 of 30 systems restored it.

EMBR – Emotional Memory for Believable Roleplay | Python, SQLite, BM25 + embeddings

2026

- Middleware giving NPCs persistent, emotion-grounded memory: hybrid BM25 and embedding retrieval, an affect-appraisal layer over a valence–arousal circumplex, and SQLite persistence behind a dependency-light core.
- Built the evaluation harness alongside it – leave-one-query-out cross-validated weight tuning applied identically to every baseline, exact permutation tests with Holm-Bonferroni correction, a 20-probe adversarial memory-injection corpus, and 115 passing tests.

TAST – Trait Activation Steering | Python, PyTorch, LLM activation engineering

2025–2026

- Per-trait activation steering for looped LLM-driven characters: steering personality expression through internal activations rather than prompt text, so persona holds across long conversations instead of drifting.
- Under revision for resubmission following AIIDE 2026 review; currently adding a prompting baseline and an audit pass over the steering-hook measurement pipeline.

PEAK – Platformer Engine for RL Benchmarking | Python, Gymnasium, Stable-Baselines3, Hydra

2025–2026

- Co-developed a deterministic high-performance DRL benchmarking engine with modular persona-based reward shaping, evaluating PPO agents across 11 custom 2D platformer stages with reproducible SMB1-style physics. GitHub: Code-SorceryLab/PEAK-DRL-Tool.
- Built a **Dual Spatial Hashing** collision system (separate static/dynamic grids) achieving $O(C)$ per-query cost and sustained **1000+ environment steps per second**, making 10M-step training runs practical on single-machine hardware.

Gestura – Gesture-Based Medical Screening | Python, Flask, CV, Gemini API, Three.js

Jan 2026

- Built at **HackHive 2026** a non-verbal patient intake tool: Flask backend streaming real-time hand-tracking state to a Three.js 3D "digital twin" frontend, with **Google Gemini** converting gestures into SOAP-style clinician summaries.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Bash, JavaScript (Node.js/Three.js), HTML/CSS, P4

ML / RL: PyTorch, Stable-Baselines3, Gymnasium, PPO, Crafter, TensorBoard, Hydra, NumPy, Pandas, Scikit-Learn

Research Methods: Experimental design, cross-validated tuning, permutation testing, bootstrap CIs, ablation studies, systematic mapping studies

AI Systems & Web: Gemini API, Ollama, LLM prompt engineering, Computer Vision, Flask, REST APIs, Docker, Git

Networking & Sec: Intel Tofino P4, Wireshark, Nmap, Cisco IOS, Linux (Kali/RedHat), Vulnerability Assessment

Certifications: Cisco Certified Network Associate (CCNA) 2024–2027 • TryHackMe 200+ Hours (2023–Present) • Dean's List 2024-2026